

Version B

Question 1:

1. What are the limitations of bidirectional RNN, where you do not use it?
2. What is the purpose of the forget gate in LSTM?
3. Which layer type is responsible for handling variable-length input in an RNN?
4. Which layer type is commonly used in RNNs for machine translation tasks?
5. What is the main difference of encoder and decoder architecture in transformer?
6. In the Transformer architecture, which is not a sequential model and uses only feed-forward networks, will the output for the input sequences
A,B,C,D,Y
Y,A,B,C,D
Will "Y" output be the same or not ? Why?

Question 2:

Ewe = [0.2,0.5,0.3,0.7]

Eare = [0.4,0.6,0.4,0.5]

Egood = [0.5,0.4,0.2,0.6]

Efriend = [0.7,0.5,0.3,0.5]

$W(q) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

$W(k) = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

$W(v) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

1. What are q,k,v
2. What is the self attention of Efriend?